

Qualification Pack



Draughtsman Mechanical

QP Code: CSC/Q0402

Version: 1.0

NSQF Level: 4

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CSC/Q0402: Draughtsman Mechanical

Brief Job Description

It involves select the appropriate equipment and drawing software to use based on the type and complexity of the drawing functions to be carried out and the use of a CAD system linked bills of material, file management and associated customization of installed software including the use of macros, menus and default settings.

Personal Attributes

Basic communication, numerical and computational abilities. Openness to learning, ability to plan and organize own work and identify and solve problems in the course of working. Understanding the need to take initiative and manage self and work to improve efficiency and effectiveness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CSC/N0402: Make or modify 2D mechanical engineering drawings using CAD system](#)
2. [CSC/N1335: Use basic health and safety practices at the workplace](#)
3. [CSC/N1336: Work effectively with others](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Machine Tools
Occupation	Design
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2004/NIL
Minimum Educational Qualification & Experience	10th Class
Minimum Level of Education for Training in School	

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Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	24/11/2017
Next Review Date	10/08/2025
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NSQC Approval Date	19/05/2015
Version	1.0
Reference code on NQR	2015/CCM/GCSC/00130
NQR Version	1.0

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CSC/N0402: Make or modify 2D mechanical engineering drawings using CAD system

Description

This unit covers the skills and knowledge needed to set up and operate a computer aided drawing (CAD) system to produce detailed drawings for engineering activities, in accordance with approved procedures.

Scope

This unit/task covers the following:

- Prepare for 2D mechanical engineering drawings
- Perform set-up activities
- Make or modify 2D mechanical engineering drawings using CAD system

Elements and Performance Criteria

Prepare for 2D mechanical engineering drawings

To be competent, the user/individual on the job must be able to:

- PC1.** use appropriate sources to obtain the technical information relevant to the drawing to be created (Technical information relevant to the drawing to be created: drawing brief; specifications (overall dimensions, materials, special procedures for manufacturing); drawing change or modification request; regulations; existing drawings/designs, sketches, notes from meetings/discussions; standards reference documents (eg. limits and fits, tapping drill charts, contraction allowances)
- PC2.** identify design features, as appropriate to the drawing being produced (Design features: function, materials, clearance, operating environment, quality, aesthetics, interfaces, physical space; tolerances)
- PC3.** ensure that the data and information received is complete and correct
- PC4.** establish the drawing requirements from the data and information received
- PC5.** report and rectify incorrect and inconsistent information in job specification documents as per organization procedures
- PC6.** access and use the correct drawing software
- PC7.** select drafting equipment appropriate to the drawing method chosen
- PC8.** check that all the equipment is correctly connected and in a safe and usable working condition
- PC9.** power up the equipment and activate the appropriate drawing software

Perform set-up activities

To be competent, the user/individual on the job must be able to:

- PC10.** customize system variables, menus and drawing defaults to produce the drawing to the appropriate scale
- PC11.** develop macros as per approved procedures

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- PC12.** set up and check that all peripheral devices are connected and correctly operating and interface with ERP if required is available (Peripheral devices could be: keyboard, mouse, light pen, digitizer/tablet, scanner, printer, plotter, etc.)
- PC13.** set the drawing datum at a convenient point
- PC14.** set up drawing parameters (eg. layers, line types, color, text styles) to company procedures or to suit the drawing produced

Make or modify 2D mechanical engineering drawings using CAD system

To be competent, the user/individual on the job must be able to:

- PC15.** interpret and produce mechanical drawings, using first angle orthographic projections, isometric/oblique projections, third angle orthographic projections, sectional views
- PC16.** apply drafting principles to produce various types of drawings that are consistent with applicable standards and procedures for use in various engineering activities (Types of drawings: detail drawings, sub-assembly drawings, general arrangement drawings, installation drawings, exploded views Standards and procedures: organizational guidelines and procedures, recognized compliance agency/body standards, directives or codes of practice, CAD software standards/protocols, national and/or International standards or directives, customer standards and requirements, health, safety and environmental requirements Engineering Activities: production activities (such as processing of materials, fabrication, finishing, assembly, joining); installation activities (such as commissioning/decommissioning, site preparation, equipment installation); operational activities (such as movement of materials, workplace layouts, work-flow diagrams), maintenance activities (such as planned preventative maintenance, part/sub-asset exchange)
- PC17.** create a drawing template to the required standards, which includes all necessary detail (eg.) using various drawing tools (Drawing template details: layers of drawings, scale, paper size, color setup, line types, dimension system, title, drawing number, date, text styles Drawing Tools: straight lines, hatching and shading on drawings, adding dimensions and text to drawings, producing layers of drawings, symbols and abbreviations, hidden detail, curved/contour lines, angled lines, circles or ellipses; parts lists, geometrical and dimensional tolerance, insertion of standard components, elevation, plane view, side view, sectional views, detail views)
- PC18.** use appropriate terminologies and techniques to create drawings, in the required formats, that are sufficiently and clearly detailed
- PC19.** use keyboard command and pull down menus available in common CAD systems
- PC20.** use codes and other references that follow the required conventions
- PC21.** draw temporary fasteners and rivets
- PC22.** draw components details and assembly drawings
- PC23.** draw piping layouts, gears and machine foundation or base
- PC24.** draw working drawings of jigs and fixtures
- PC25.** draw detailed drawings of dies, moulds and press tools
- PC26.** dimension and label the drawing as per approved procedures
- PC27.** create detailed views using various scales to meet job requirements
- PC28.** ensure that drawings are checked and approved by the appropriate person
- PC29.** produce hard copies of the finished drawings

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- PC30.** check that the drawing is correctly titled and referenced; sawing is correctly titled and referenced
- PC31.** save the drawing to an appropriate storage medium (eg. hard drive, CD/DVD, external storage device)
- PC32.** create a separate backup copy and place it in safe storage
- PC33.** identify component parts list with part name, description of part, material specification or part number, quantities and other details to prepare bill of materials as per organizational guidelines
- PC34.** deal promptly and effectively with problems within control and seek help and guidance from the relevant people if you have problems that they cannot resolve
- PC35.** ensure that changes are completed as required by organizational procedures
- PC36.** shut down the CAD system to a safe condition on completion of the drawing activities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** escalation matrix and procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** organizational procedures and information systems for retrieving and storing drawing data
- KU11.** system variables that can be customized
- KU12.** procedures and need for customizing identified system variables
- KU13.** applicable drafting standards/procedures
- KU14.** procedures and need for customizing menus and system defaults
- KU15.** procedures and need for developing macros
- KU16.** appropriate projection for the drawing purpose
- KU17.** reasons for selecting the chosen projection
- KU18.** reasons for including auxiliary views in drawings
- KU19.** procedures for producing component, layout and/or assembly drawings
- KU20.** drawing specifications
- KU21.** common symbols used in drawings
- KU22.** how and where to obtain the relevant sources and methods for obtaining any required technical information relevant to the drawing

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- KU23.** methods and procedures used to minimize the chances of infecting a computer with a virus
- KU24.** procedure to follow in case there are corruptions or virus attacks
- KU25.** practices that make systems vulnerable to corruption and damage
- KU26.** basic set-up and operation of the computer system, and the peripheral devices that are used (eg. light pen, digitizer and tablet, printer or plotter, scanner)
- KU27.** how to access the specific computer drawing software to be used, and the use of software manuals and related documents to aid operation of the relevant drawing system
- KU28.** basic principles of engineering manufacturing operations that are used to produce the drawn item (Basic principles of engineering manufacturing operations: casting and forging; fabrication; machining methods; joining processes; assembly and installation methods; limitations of the equipment/processes; kinematics principles relevant to manufacturing of machinery)
- KU29.** types of drawings that may be produced by the software
- KU30.** selection of standard components
- KU31.** functionality of the component being drawn, and its interrelationship with other components and assemblies
- KU32.** how to set up the viewing screen to show multiple views of the drawing to help with drawing creation
- KU33.** standards and conventions that are used for the drawings
- KU34.** how to set up the drawing template parameters
- KU35.** application and use of drawing tools
- KU36.** how to access, recognize and use a wide range of standard components and symbol libraries from the CAD equipment
- KU37.** need for document control
- KU38.** how to save and store drawings
- KU39.** need to create backup copies, and to file them in a separate and safe location
- KU40.** how to produce hard copies of the drawings, and the advantages and disadvantages of printers and plotters

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- GS3.** undertake numerical operations, and calculations/ formulae (Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages)
- GS4.** identify and draw various basic, compound and solid shapes as per dimensions given (Basic shapes: square, rectangle, triangle, circle Compound shapes: involving squares, rectangles, triangles, circles, semi-circles, quadrants of a circle Solid shapes: cube, rectangular prism, cylinder)

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- GS5.** use appropriate units and number systems to express degree of accuracy (Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity)
- GS6.** interpret and express tolerance in terms of limits on dimensions
- GS7.** calculation of the value of angles in a triangle (Angles in a triangle: right-angled, isosceles, equilateral)
- GS8.** convey and share technical information clearly using appropriate language
- GS9.** check and clarify task-related information
- GS10.** liaise with appropriate authorities using correct protocol
- GS11.** communicate with people in respectful form and manner in line with organizational protocol
- GS12.** plan, prioritize and sequence work operations as per job requirements
- GS13.** organize and analyze information relevant to work
- GS14.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS15.** exercise restraint while expressing dissent and during conflict situations
- GS16.** avoid and manage distractions to be disciplined at work
- GS17.** manage own time for achieving better results
- GS18.** work in a team in order to achieve better results
- GS19.** identify and clarify work roles within a team
- GS20.** communicate and cooperate with others in the team for better results
- GS21.** seek assistance from fellow team members
- GS22.** identify problems with work planning, procedures, output and behavior and their implications
- GS23.** prioritize and plan for problem solving
- GS24.** communicate problems appropriately to others
- GS25.** identify sources of information and support for problem solving
- GS26.** seek assistance and support from other sources to solve problems
- GS27.** identify effective resolution techniques
- GS28.** select and apply resolution techniques
- GS29.** seek evidence for problem resolution
- GS30.** undertake and express new ideas and initiatives to others
- GS31.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS32.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS33.** enhance ones competencies in new and different situations and contexts to achieve more
- GS34.** participate in on-the-job and other learning, training and development interventions and assessments
- GS35.** clarify task related information with appropriate personnel or technical adviser
- GS36.** seek to improve and modify own work practices
- GS37.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for 2D mechanical engineering drawings</i>	6	15	-	-
PC1. use appropriate sources to obtain the technical information relevant to the drawing to be created (Technical information relevant to the drawing to be created: drawing brief; specifications (overall dimensions, materials, special procedures for manufacturing); drawing change or modification request; regulations; existing drawings/designs, sketches, notes from meetings/discussions; standards reference documents (eg. limits and fits, tapping drill charts, contraction allowances)	-	2	-	-
PC2. identify design features, as appropriate to the drawing being produced (Design features: function, materials, clearance, operating environment, quality, aesthetics, interfaces, physical space; tolerances)	2	2	-	-
PC3. ensure that the data and information received is complete and correct	-	2	-	-
PC4. establish the drawing requirements from the data and information received	1	2	-	-
PC5. report and rectify incorrect and inconsistent information in job specification documents as per organization procedures	1	2	-	-
PC6. access and use the correct drawing software	1	1	-	-
PC7. select drafting equipment appropriate to the drawing method chosen	1	2	-	-
PC8. check that all the equipment is correctly connected and in a safe and usable working condition	-	1	-	-
PC9. power up the equipment and activate the appropriate drawing software	-	1	-	-
<i>Perform set-up activities</i>	4	10	-	-
PC10. customize system variables, menus and drawing defaults to produce the drawing to the appropriate scale	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. develop macros as per approved procedures	2	2	-	-
PC12. set up and check that all peripheral devices are connected and correctly operating and interface with ERP if required is available (Peripheral devices could be: keyboard, mouse, light pen, digitizer/tablet, scanner, printer, plotter, etc.)	-	2	-	-
PC13. set the drawing datum at a convenient point	-	2	-	-
PC14. set up drawing parameters (eg. layers, line types, color, text styles) to company procedures or to suit the drawing produced	1	2	-	-
<i>Make or modify 2D mechanical engineering drawings using CAD system</i>	20	45	-	-
PC15. interpret and produce mechanical drawings, using first angle orthographic projections, isometric/oblique projections, third angle orthographic projections, sectional views	2	3	-	-
PC16. apply drafting principles to produce various types of drawings that are consistent with applicable standards and procedures for use in various engineering activities (Types of drawings: detail drawings, sub-assembly drawings, general arrangement drawings, installation drawings, exploded views Standards and procedures: organizational guidelines and procedures, recognized compliance agency/body standards, directives or codes of practice, CAD software standards/protocols, national and/or International standards or directives, customer standards and requirements, health, safety and environmental requirements Engineering Activities: production activities (such as processing of materials, fabrication, finishing, assembly, joining); installation activities (such as commissioning/decommissioning, site preparation, equipment installation); operational activities (such as movement of materials, workplace layouts, work-flow diagrams), maintenance activities (such as planned preventative maintenance, part/sub-asset exchange)	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. create a drawing template to the required standards, which includes all necessary detail (eg.) using various drawing tools (Drawing template details: layers of drawings, scale, paper size, color setup, line types, dimension system, title, drawing number, date, text styles Drawing Tools: straight lines, hatching and shading on drawings, adding dimensions and text to drawings, producing layers of drawings, symbols and abbreviations, hidden detail, curved/contour lines, angled lines, circles or ellipses; parts lists, geometrical and dimensional tolerance, insertion of standard components, elevation, plane view, side view, sectional views, detail views)	2	3	-	-
PC18. use appropriate terminologies and techniques to create drawings, in the required formats, that are sufficiently and clearly detailed	2	2	-	-
PC19. use keyboard command and pull down menus available in common CAD systems	1	1	-	-
PC20. use codes and other references that follow the required conventions	1	2	-	-
PC21. draw temporary fasteners and rivets	1	2	-	-
PC22. draw components details and assembly drawings	1	3	-	-
PC23. draw piping layouts, gears and machine foundation or base	1	3	-	-
PC24. draw working drawings of jigs and fixtures	1	3	-	-
PC25. draw detailed drawings of dies, moulds and press tools	1	3	-	-
PC26. dimension and label the drawing as per approved procedures	1	3	-	-
PC27. create detailed views using various scales to meet job requirements	1	2	-	-
PC28. ensure that drawings are checked and approved by the appropriate person	-	1	-	-
PC29. produce hard copies of the finished drawings	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC30. check that the drawing is correctly titled and referenced; sawing is correctly titled and referenced	-	2	-	-
PC31. save the drawing to an appropriate storage medium (eg. hard drive, CD/DVD, external storage device)	-	1	-	-
PC32. create a separate backup copy and place it in safe storage	-	1	-	-
PC33. identify component parts list with part name, description of part, material specification or part number, quantities and other details to prepare bill of materials as per organizational guidelines	2	2	-	-
PC34. deal promptly and effectively with problems within control and seek help and guidance from the relevant people if you have problems that they cannot resolve	-	2	-	-
PC35. ensure that changes are completed as required by organizational procedures	1	1	-	-
PC36. shut down the CAD system to a safe condition on completion of the drawing activities	-	1	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0402
NOS Name	Make or modify 2D mechanical engineering drawings using CAD system
Sector	Capital Goods
Sub-Sector	Machine Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Design
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/11/2017
Next Review Date	31/03/2022
NSQC Clearance Date	19/05/2015

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CSC/N1335: Use basic health and safety practices at the workplace

Description

This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment.

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuffless (without folds), trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors Equipment: hand shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** identify job-site hazardous work and state possible causes of risk or accident in the workplace (Hazards: sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces(sharp, slippery, uneven, chipped, broken, etc.); hazardous substances(chemicals, gas, oxy-fuel, fumes, dust, etc.); physical hazards(working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc.) electrical hazards (power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.) Possible causes of risk and accident: physical actions; reading; listening to and giving instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness))
- PC5.** carry out safe working practices while dealing with hazards to ensure the safety of self and others (Safe working practices: using protective clothing and equipment; putting up and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter, spillage and unsafe object lying casually; while working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc.; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working in confined places, trenches or at heights, etc. including safety harness, fall arrestors, etc.)
- PC6.** state methods of accident prevention in the work environment of the job role (Methods of accident prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices (such as safe carrying procedures); safety notices, advice; instruction from colleagues and supervisors)
- PC7.** state location of general health and safety equipment in the workplace (General health and safety equipment: fire extinguishers; first aid equipment; safety instruments and clothing; safety installations(eg fire exits, exhaust fans))

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- PC8.** inspect for faults, set up and safely use steps and ladders in general use (Ladder faults: corrosion of metal components, deterioration, splits and cracks timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.) (Ladders set up: firm/level base, clip/lash down, leaning at the correct angle, etc.)
- PC9.** work safely in and around trenches, elevated places and confined areas
- PC10.** lift heavy objects safely using correct procedures
- PC11.** apply good housekeeping practices at all times (Good housekeeping practices: clean/tidy work areas, removal/disposal of waste products, protect surfaces)
- PC12.** identify common hazard signs displayed in various areas (Various areas: on chemical containers; equipment; packages; inside buildings; in open areas and public spaces, etc.)
- PC13.** retrieve and/or point out documents that refer to health and safety in the workplace (Documents: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (eg government notices))

Fire safety

To be competent, the user/individual on the job must be able to:

- PC14.** use the various appropriate fire extinguishers on different types of fires correctly (Types of fires: Class A: eg. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.; Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: eg. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity); Class D: combustible metals such as magnesium, titanium, and sodium (These fires burn at extremely high temperatures and require special suppression agents))
- PC15.** demonstrate rescue techniques applied during fire hazard
- PC16.** demonstrate good housekeeping in order to prevent fire hazards
- PC17.** demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC18.** demonstrate how to free a person from electrocution
- PC19.** administer appropriate first aid to victims where required eg. in case of bleeding, burns, choking, electric shock, poisoning etc.
- PC20.** demonstrate basic techniques of bandaging
- PC21.** respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments
- PC22.** perform and organize loss minimization or rescue activity during an accident in real or simulated environments
- PC23.** administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases
- PC24.** demonstrate the artificial respiration and the CPR Process
- PC25.** participate in emergency procedures (Emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work)
- PC26.** complete a written accident/incident report or dictate a report to another person, and send report to person responsible (Incident Report includes details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified)

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Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC27. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
- KU2.** names and location of documents that refer to health and safety in the workplace
- KU3.** meaning of hazards and risks
- KU4.** health and safety hazards commonly present in the work environment and related precautions
- KU5.** possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU6.** possible causes of risk and accident(Possible causes of risk and accident: physical actions; reading; listening to and giving instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness))
- KU7.** methods of accident prevention(Methods of accident prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices (such as safe carrying procedures); safety notices, advice; instruction from colleagues and supervisors)
- KU8.** safe working practices when working with tools and machines
- KU9.** safe working practices while working at various hazardous sites
- KU10.** where to find all the general health and safety equipment in the workplace
- KU11.** various dangers associated with the use of electrical equipment
- KU12.** preventative and remedial actions to be taken in the case of exposure to toxic materials(Exposure: ingested, contact with skin, inhaled); (Preventative action: ventilation, masks, protective clothing/ equipment); (Remedial action: immediate first aid, report to supervisor Toxic materials: solvents, flux, lead)
- KU13.** importance of using protective clothing/equipment while working
- KU14.** precautionary activities to prevent the fire accident
- KU15.** various causes of fire(Causes of fires: heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.)
- KU16.** techniques of using the different fire extinguishers
- KU17.** different methods of extinguishing fire
- KU18.** different materials used for extinguishing fire(Materials: sand, water, foam, CO2, dry powder)
- KU19.** rescue techniques applied during a fire hazard
- KU20.** various types of safety signs and what they mean
- KU21.** appropriate basic first aid treatment relevant to the condition eg. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU22.** content of written accident report
- KU23.** potential injuries and ill health associated with incorrect manual handling

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KU24. safe lifting and carrying practices

KU25. personal safety, health and dignity issues relating to the movement of a person by others

KU26. potential impact to a person who is moved incorrectly

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and comprehend basic content to read labels, charts, signages

GS2. read and comprehend basic English to read manuals of operations

GS3. read an accident/incident report in local language or English

GS4. write an accident/incident report in local language or English

GS5. question co-workers appropriately in order to clarify instructions and other issues

GS6. give clear instructions to coworkers, subordinates others

GS7. make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines

GS8. plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity

GS9. remain congenial while discussing and debating issues with co-workers

GS10. follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice

GS11. ask for, provide and receive required assistance where possible to ensure achievement of work related objectives

GS12. thank co-workers for any assistance received

GS13. offer appropriate respect based on mutuality and respect for fellow workmanship and authority

GS14. identify immediate or temporary solutions to resolve delays

GS15. identify sources of support that can be availed of for problem solving for various kind of problems

GS16. seek appropriate assistance from other sources to resolve problems

GS17. report problems that you cannot resolve to appropriate authority

GS18. identify cause and effect relations in their area of work

GS19. use cause and effect relations to anticipate potential problems and their solution

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	21	32	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuffless (without folds), trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors Equipment: hand shields, machine guards, residual current devices, shields, dust sheets, respirator)	1	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	1	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	1	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace (Hazards: sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces(sharp, slippery, uneven, chipped, broken, etc.); hazardous substances(chemicals, gas, oxy-fuel, fumes, dust, etc.); physical hazards(working at heights, large and heavy objects and machines, sharp and piercing objects, tools and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc.) electrical hazards (power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.) Possible causes of risk and accident: physical actions; reading; listening to and giving instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness))	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. carry out safe working practices while dealing with hazards to ensure the safety of self and others (Safe working practices: using protective clothing and equipment; putting up and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter, spillage and unsafe object lying casually; while working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc.; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working in confined places, trenches or at heights, etc. including safety harness, fall arrestors, etc.)	2	2	-	-
PC6. state methods of accident prevention in the work environment of the job role (Methods of accident prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices (such as safe carrying procedures); safety notices, advice; instruction from colleagues and supervisors)	2	1	-	-
PC7. state location of general health and safety equipment in the workplace (General health and safety equipment: fire extinguishers; first aid equipment; safety instruments and clothing; safety installations(eg fire exits, exhaust fans))	2	3	-	-
PC8. inspect for faults, set up and safely use steps and ladders in general use (Ladder faults: corrosion of metal components, deterioration, splits and cracks timber components, imbalance, loose rungs, missing/unfixed nuts or bolts, etc.) (Ladders set up: firm/level base, clip/lash down, leaning at the correct angle, etc.)	2	3	-	-
PC9. work safely in and around trenches, elevated places and confined areas	2	3	-	-
PC10. lift heavy objects safely using correct procedures	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. apply good housekeeping practices at all times (Good housekeeping practices: clean/tidy work areas, removal/disposal of waste products, protect surfaces)	2	3	-	-
PC12. identify common hazard signs displayed in various areas (Various areas: on chemical containers; equipment; packages; inside buildings; in open areas and public spaces, etc.)	1	2	-	-
PC13. retrieve and/or point out documents that refer to health and safety in the workplace (Documents: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (eg government notices))	1	3	-	-
<i>Fire safety</i>	4	10	-	-
PC14. use the various appropriate fire extinguishers on different types of fires correctly (Types of fires: Class A: eg. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.; Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: eg. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity); Class D: combustible metals such as magnesium, titanium, and sodium (These fires burn at extremely high temperatures and require special suppression agents))	1	2	-	-
PC15. demonstrate rescue techniques applied during fire hazard	1	2	-	-
PC16. demonstrate good housekeeping in order to prevent fire hazards	1	3	-	-
PC17. demonstrate the correct use of a fire extinguisher	1	3	-	-
<i>Emergencies, rescue and first-aid procedures</i>	9	20	-	-
PC18. demonstrate how to free a person from electrocution	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. administer appropriate first aid to victims where required eg. in case of bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC20. demonstrate basic techniques of bandaging	1	2	-	-
PC21. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC22. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC23. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC24. demonstrate the artificial respiration and the CPR Process	1	2	-	-
PC25. participate in emergency procedures (Emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work)	1	3	-	-
PC26. complete a written accident/incident report or dictate a report to another person, and send report to person responsible (Incident Report includes details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified)	1	2	-	-
<i>Emergencies, rescue and first-aid procedures</i>	2	2	-	-
PC27. demonstrate correct method to move injured people and others during an emergency	2	2	-	-
NOS Total	36	64	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N1335
NOS Name	Use basic health and safety practices at the workplace
Sector	Capital Goods
Sub-Sector	Machine Tools, Process Plant Machinery, Dies, Moulds and Press Tools, Electrical and Power Machinery, Plastics Manufacturing Machinery, Light Engineering Goods, Textile Manufacturing Machinery
Occupation	Machining
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

CSC/N1336: Work effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening etc.

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working (Communication etiquette: do not use abusive language; use appropriate titles and terms of respect; do not eat or chew while talking (vice versa) etc.)
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace (Disciplined behaviors: e.g. punctuality; completing tasks as per given time and standards; not gossiping and idling time; eliminating waste, honesty, etc.)
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and coordinate within the organization
- KU6.** importance of effective communication in the workplace

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- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication
- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read basic terms and terminologies to accurately interpret work-related documents, labels, supervisor instructions in the local language
- GS2.** read and interpret accurate information from various relevant work instructions and records
- GS3.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS4.** write basic numbers, quantities and work-related terminology for operational requirements in the local language
- GS5.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans, and other associated requirements
- GS6.** give clear instructions to co-workers about the type of output required and answer queries
- GS7.** display active listening skills while interacting with co-workers and others in the workplace
- GS8.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS9.** take steps within ones limits of authority to initiate modification in plan if the circumstances require it
- GS10.** check that work meets customer requirements
- GS11.** deliver consistent and reliable service to internal and external customers
- GS12.** work with co-workers and supervisor to resolve any issues that threaten disruption, increase risk, cause delays or under-achievement of quality and targets as per the planned schedule

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working (Communication etiquette: do not use abusive language; use appropriate titles and terms of respect; do not eat or chew while talking (vice versa) etc.)	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace (Disciplined behaviors: e.g. punctuality; completing tasks as per given time and standards; not gossiping and idling time; eliminating waste, honesty, etc.)	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N1336
NOS Name	Work effectively with others
Sector	Capital Goods
Sub-Sector	Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Machining
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	30/09/2021
Next Review Date	30/09/2024
NSQC Clearance Date	30/09/2021

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for the Qualification Pack will be created by CGSC.
2. Performance Criteria (PC) have been assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
3. The assessment for the theory part will/may be based on knowledge bank of questions approved CGSC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. Assessment Agencies will create Assessor Guides comprising of Theory and Practical Assessment Set and Guidelines for each examination/training centre (as per assessment criteria below). The same will be approved by CGSC for adequacy.
6. To successfully attain Certification on the Qualification Pack, the trainee must score a minimum of 70% in each Core NOS and minimum of 50% in all non-core NOS. In addition, a candidate needs to attain a minimum overall pass percentage of 70% for certification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CSC/N0402.Make or modify 2D mechanical engineering drawings using CAD system	30	70	-	-	100	80
CSC/N1335.Use basic health and safety practices at the workplace	36	64	-	-	100	10
CSC/N1336.Work effectively with others	30	70	-	-	100	10
Total	96	204	-	-	300	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
CNC	Computer Numerically Controlled
CAD	Computer Aided Design
2D	2 Dimensional
3D	3 Dimensional
CO₂	Carbon Dioxide
CPR	Cardiac Pulmonary Resuscitation
ISO	International Organization For Standardization
PPE	Personal Protective Equipment
CD	Compact Disc
DVD	Digital Video Disc Or Digital Versatile Disc

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.